

# AETHERIA: THE HUNTER'S PROTOCOL

Full-Stack 'Life RPG' Hackathon Blueprint, Game Architecture & Winning Strategy

TARGET: Web Hackathon 2026 | STACK: Next.js 15, PostgreSQL, Framer Motion | PERSISTENCE: 100% Cloud DB

**CRITICAL ZERO-TOLERANCE WARNING:** Submissions relying solely on *localStorage* for data persistence, having broken/inaccessible live links, crashing during evaluation, having fewer than 3 chronological git commits, or omitting the 90–180s walkthrough video (<100MB) receive an **AUTOMATIC ZERO**. Our architecture strictly eliminates all 6 disqualification vectors.

## 1. Core Problem & Creative Direction

Standard productivity tools and habit trackers fail because real habits suffer from **delayed gratification** (hitting the gym, reading, or coding takes months to see results). Video games hook players through **instant feedback loops, clear progression systems, and tangible rewards**. Our mission is to build a full-stack Life RPG application bridging this gap.

**Thematic Concept: 'Solo Leveling / Dark Fantasy Hunter Guild'** — Judges explicitly warned that generic SaaS or unstyled CRUD apps will receive low marks. We implement:

- Cohesive RPG Vocabulary:** Tasks → *Quests & Bounties (Rank E to S)*; Points → *Gold & Mana Crystals*; Habits → *Daily Rites & Streaks*; Store → *Guild Black Market*.
- Alive & Tactile Feedback:** Built-in retro audio SFX (blade swing on complete, coin chime, fanfare on level-up with mute toggle). Floating +XP/+Gold spring physics flying to top navigation.
- Celebratory Milestone Modals:** Screen dimming, spinning animated hunter badge, and particle confetti burst via canvas-confetti.
- Zero-Lag Native Feel:** Optimistic UI updates (checkmarks complete in 0ms; backend syncs asynchronously) and skeleton screens.

## 2. Technical Architecture & Tech Stack

| Layer          | Technology Choice                             | Evaluation Advantage / Value                                                                            |
|----------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Framework      | Next.js 15 (App Router, React 19, TypeScript) | Unified frontend & backend API routes, fast SSR, zero CORS issues, instant Vercel deploy.               |
| UI & Styling   | Tailwind CSS + shadcn/ui + Lucide Icons       | Dark mode by default, accessible Radix primitives with full keyboard navigation (Tab/Enter).            |
| Animations     | Framer Motion + Canvas-Confetti               | Spring micro-interactions, celebratory level-up popups, floating XP feedback.                           |
| Database & ORM | PostgreSQL (Supabase / Neon Cloud) + Prisma   | <b>100% Real DB Persistence</b> across hard refreshes and cross-device sync. Relational user isolation. |
| Audio Engine   | Web Audio API / Howler.js                     | Zero-latency retro game audio effects with header mute toggle switch.                                   |
| Deployment     | Vercel (App) + Supabase (Database)            | Public HTTPS live URL required for hackathon submission with 99.9% uptime.                              |

### 3. RPG Progression Engine & Game Mathematics

**Non-Linear Leveling Formula:** Each subsequent level requires progressively more XP:  $XP\_Req(Level) = \text{Math.floor}(100 \times Level^{1.5})$

| Level Jump   | XP Needed | Cumulative XP | Psychological Motivation                                   |
|--------------|-----------|---------------|------------------------------------------------------------|
| Level 1 → 2  | 100 XP    | 100 XP        | Instant early gratification within first 2 completed tasks |
| Level 2 → 3  | 283 XP    | 383 XP        | Requires completing a moderate quest chain                 |
| Level 5 → 6  | 1,118 XP  | 3,089 XP      | Established daily routine threshold                        |
| Level 9 → 10 | 2,700 XP  | 11,400 XP     | Prestige Hunter status / elite milestone                   |

**The 5 Core Hunter Attributes:**

- **Strength (STR):** Gym, workouts, weightlifting, physical sports.
- **Intellect (INT):** Coding, technical studying, reading textbooks, problem solving.
- **Vitality (VIT):** Sleep consistency, hydration (8 glasses), meal prepping, meditation.
- **Agility (AGI):** Speed chores, inbox zero, swift errands, tidying workspace.
- **Charisma (CHA):** Team standups, public speaking, networking, attending meetups.

### 4. Hackathon-Winning 'Killer Features' (The Wow-Factor)

|                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                       |
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| <b>1. Weekly Boss Raid: 'The Procrastination Demon'</b><br>A weekly boss with 3,000 HP (e.g. <i>Malakor the Sloth</i> ). Every completed quest strikes the boss with attack damage equal to the quest's XP output + user Strength bonus. Slaying the boss before the countdown drops rare loot chests and guild badges. | <b>2. Guild Black Market &amp; Real-World Rewards</b><br>Spend earned Gold on: (a) <i>Virtual Relics</i> : Scroll of Focus (+10% INT XP), Aegis Shield (Streak Freeze); and (b) <i>Real-Life Custom Rewards</i> : Users set up personal rewards ('1 hr Gaming' = 100 Gold, 'Order Pizza' = 350 Gold) bridging game loot to real life! |
| <b>3. Hunter Rank Difficulty Tiers (E → S)</b><br>Quests have difficulty ratings: E-Rank (10 XP, 5 Gold), C-Rank (35 XP, 15 Gold), A/S-Rank Boss Milestones (150+ XP, 80 Gold). Visual rank badges mirror Solo Leveling hunter licenses.                                                                                | <b>4. Accessibility &amp; Command Palette (Ctrl+K)</b><br>Fully compliant with accessibility guidelines. Press Ctrl+K or C for instant quest modal; navigate quests via Tab, mark complete via Space. 100% responsive bottom-bar on mobile devices.                                                                                   |

### 5. Database Schema & Data Integrity Model



## 6. Zero-Tolerance Disqualification Compliance Matrix

| Rule                  | Official Disqualification Trigger                    | Our Absolute Prevention Countermeasure                                                          |
|-----------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Broken Links          | Private GitHub repo or 404 live link at judging.     | Verify public permissions on repo; Vercel deployment with SSL and custom health check endpoint. |
| Fake Persistence      | App relies solely on localStorage.                   | All mutations write to PostgreSQL. State proven preserved upon hard refresh (Ctrl+F5).          |
| Build/Deploy Failure  | App crashes on load in production deployment.        | Strict TypeScript checks and local production build test (npm run build) prior to git push.     |
| Console/Runtime Crash | Unhandled exceptions or blank-screen crashes.        | React Error Boundaries on all routes; comprehensive Zod schema validation on all API endpoints. |
| Invalid Repository    | Fewer than 3 commits, missing backend, code dump.    | Maintain 15+ clean chronological commits following conventional commit specifications.          |
| Video Faults          | Missing, requires login, >100MB, or outside 90-180s. | Strict 120s walkthrough video, encoded as ~25MB MP4, public unlisted YouTube + in repo.         |

## 7. Walkthrough Video Script (Strictly 120 Seconds / Under 100MB)

- 0:00 - 0:20 (Authentication & Status):** Show signup/login screen. Click 'Demo Hunter' button for instant entry. Highlight character status bar (Level 1 Novice, 0 XP, 50 Starter Gold).
- 0:20 - 0:50 (Add & Complete Quest):** Create a new B-Rank Quest ('Build Auth API', category: Intellect). Mark it complete: show floating +75 XP, audio chime, and boss taking damage.
- 0:50 - 1:15 (Level Up Celebration):** Complete second quest to cross the XP threshold. Trigger confetti, sound fanfare, and animated Level Up modal showing stat boosts.
- 1:15 - 1:35 (Persistence Proof):** **Perform a hard browser refresh (Ctrl+F5).** Explicitly show that XP, Level, Gold, and completed status remain intact from PostgreSQL.
- 1:35 - 2:00 (Shop & Boss Raid):** Visit Guild Market, purchase a Relic with earned Gold, show the Boss HP bar, and demonstrate responsive mobile view.

## 8. Suggested Team Work Distribution (Editable)

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| <b>Teammate A: Frontend, Theming &amp; UI</b> <ul style="list-style-type: none"><li>Dark theme palette, typography, layout</li><li>Framer Motion spring physics &amp; confetti modal</li><li>Retro audio effects integration with sound toggle</li><li>Mobile responsive navigation &amp; command palette</li></ul> | <b>Teammate B: Backend, Database &amp; Game Logic</b> <ul style="list-style-type: none"><li>PostgreSQL schema setup (Prisma / Supabase)</li><li>Auth session management &amp; Quest CRUD endpoints</li><li>Server-side non-linear XP &amp; streak validation</li><li>Boss raid damage calculation &amp; shop transaction API</li></ul> |
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